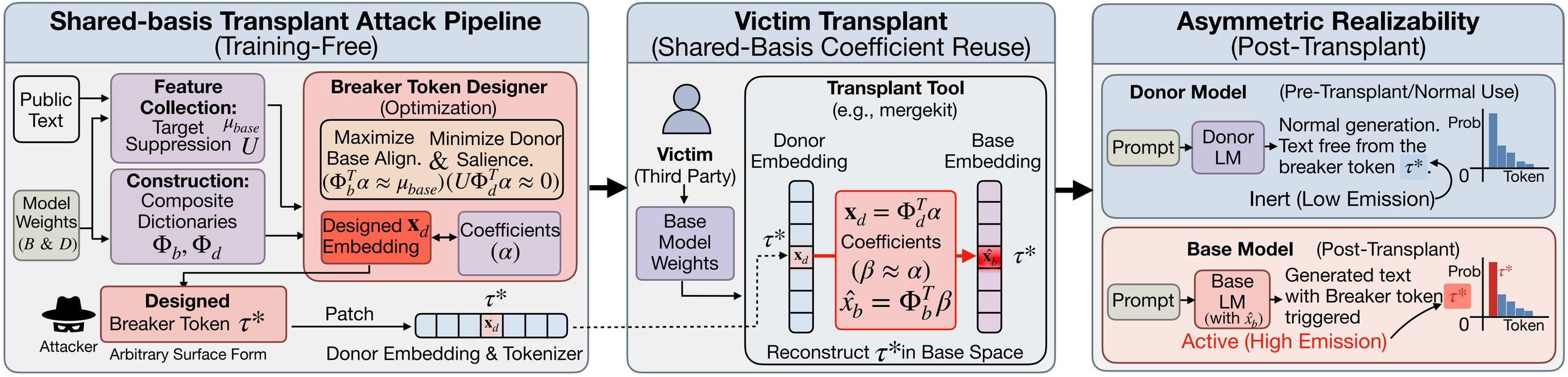
Shared-basis Transplant Attack Pipeline
(Training-Free)
Public Text
Model Weights
(B & D)
Feature Collection:
Target Suppression
μ_{base}
U
Construction:
Composite Dictionaries
Φ_b, Φ_d
Breaker Token Designer
(Optimization)
Maximize Base Align. & Minimize Donor Salience.
$(\Phi_b^T\alpha \approx \mu_{base})(U\Phi_d^T\alpha \approx 0)$
Designed $\mathbf{x}_d$ Embedding
Coefficients (α)
Attacker
Designed Breaker Token τ^*
Arbitrary Surface Form
Patch
τ^*
$\mathbf{x}_d$
Donor Embedding & Tokenizer
Victim Transplant
(Shared-Basis Coefficient Reuse)
Victim
(Third Party)
Base Model Weights
Transplant Tool
(e.g., mergekit)
Donor Embedding
Base Embedding
τ^*
$\mathbf{x}_d$
$\mathbf{x}_d = \Phi_d^T\alpha$
Coefficients
$(\beta \approx \alpha)$
$\hat{x}_b = \Phi_b^T\beta$
$\hat{\mathbf{x}}_b$
τ^*
Reconstruct τ^*in Base Space
Asymmetric Realizability
(Post-Transplant)
Donor Model
(Pre-Transplant/Normal Use)
Prompt
Donor LM
Normal generation. Text free from the breaker token τ^*.
Inert (Low Emission)
Prob
0
Token
Base Model
(Post-Transplant)
Prompt
Base LM
(with $\hat{x}_b$)
Generated text with Breaker token triggered
Active (High Emission)
Prob
τ^*
0
Token